

DIGITAL DESIGN & COMPUTER ARCHITECTURE (DDCA) PROJECT

Design & Simulation of a Smart Parking Slot Availability and Entry Control System

Hardware Architecture, Finite State Machine (FSM) Logic, and Verilog RTL Verification



Target: FPGA Synthesis



HDL: Verilog



Logic: Synchronous FSM

Problem Statement & Objectives

Replacing manual parking oversight with automated hardware logic

⚠ Urban Parking Challenges

Traffic Congestion: Time wasted searching for vacant slots.

Manual Errors: Inaccurate vehicle count tracking.

Security Gaps: Unregulated access at entry barriers.

✅ Hardware System Objectives

Real-Time Tracking: Automated sensor-based slot counting.

FSM Entry Gate: Passcode-verified gate motor trigger.

Visual Indicators: 7-Segment display output for drivers.

HARDWARE BREAKDOWN

Top-Level System Architecture

Modular hardware hierarchy designed for FPGA deployment



IR Sensor Interface

Detects vehicle arrival and exit signals.



Main FSM Logic

Controls state transitions and authentication.



Up/Down Counter

Maintains total occupancy and free slot counts.



7-Segment Decoder

Drives HEX displays for real-time status.

Finite State Machine (FSM) Specification

Deterministic state progression for secure entry and exit

 FSM Diagram

1. IDLE State

System waits for entry sensor signal.

2. WAIT_AUTH State

Evaluates user password inputs.

3. GATE_OPEN State

Lifts barrier and updates counter.

4. FULL State

Blocks entry and triggers RED LED.

Synchronous Counter Logic

Bidirectional tracking with boundary condition enforcement

```
always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        occupied_count <= 4'b0000;
    end else begin
        case ({sensor_enter, sensor_exit})
            2'b10: if (!is_full)
                    occupied_count <= occupied_count + 1;
            2'b01: if (occupied_count > 0)
                    occupied_count <= occupied_count - 1;
            default: occupied_count <= occupied_count;
        endcase
    end
end
```

Overflow Guard

Inhibits entry increment when capacity reaches max limit.

Underflow Safety

Prevents negative counter underflow on exit signals.

Synchronous Reset

Active-low reset clears system count instantly.

Entry Barrier & Security Control

Password verification and physical motor actuation interface



Auth Comparator

Combinational logic checks input key against secret bits.



Barrier Motor Control

Generates pulse signal to rotate gate motor open/close.



Status LED Drivers

Green LED for grant; Red LED and alarm for denied access.

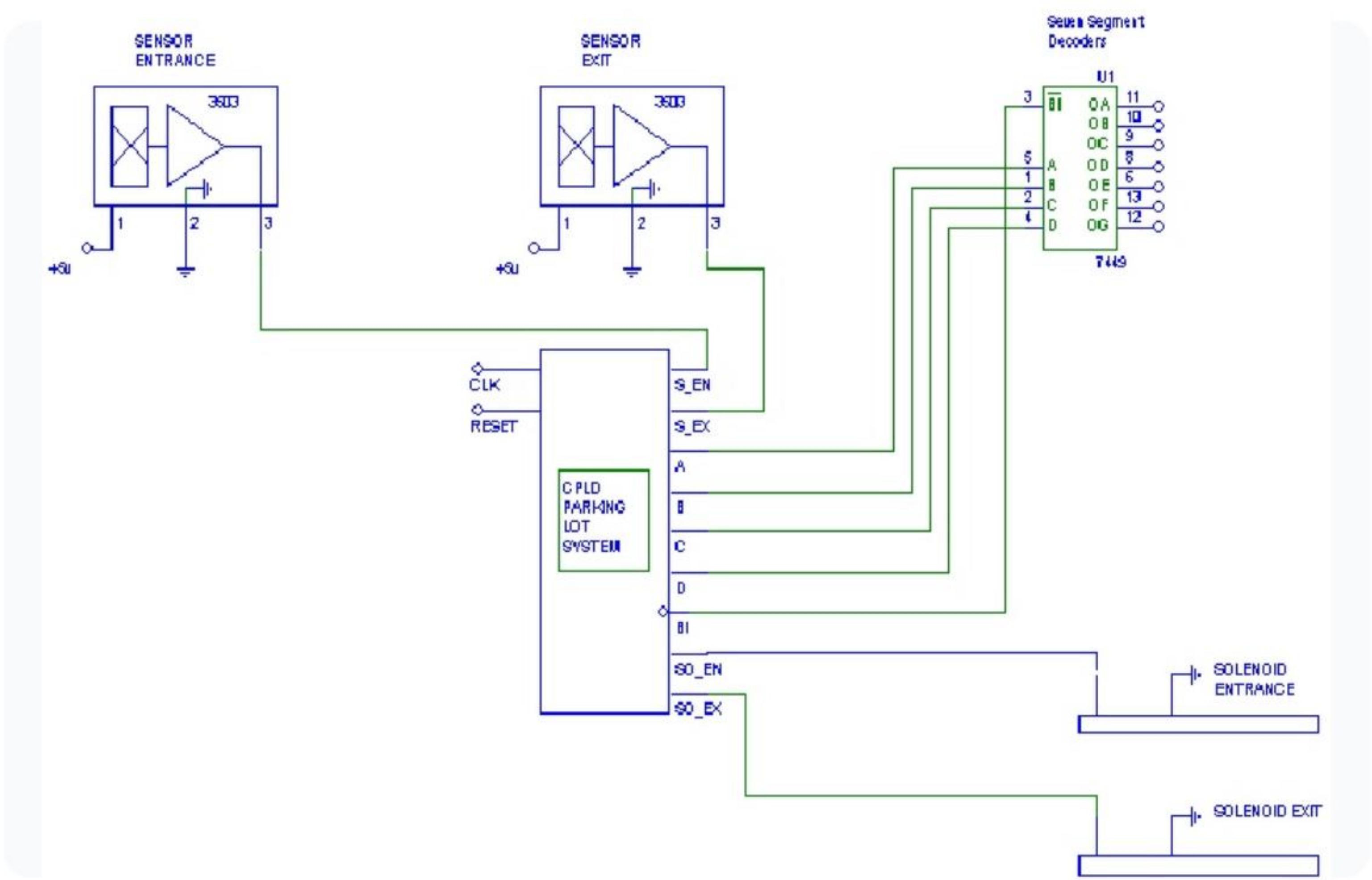
7-Segment Display & Driver Modules

Binary-to-BCD conversion for visual slot information

Display Metric	Logic Function	Decoder Output	Hardware Target
Occupied Slots	Direct Counter Bus	7-Segment Cathode Array	Display HEX_1
Available Slots	Subtractor (MAX - Count)	7-Segment Cathode Array	Display HEX_2
FULL Warning	AND Gate Comparator	Blinking / Alarm Output	Buzzer & Red LED

RTL Synthesis & Resource Utilization

Gate-level mapping and FPGA area report



32

LUTs Consumed

12

Registers / FF

0

Latches Inferred

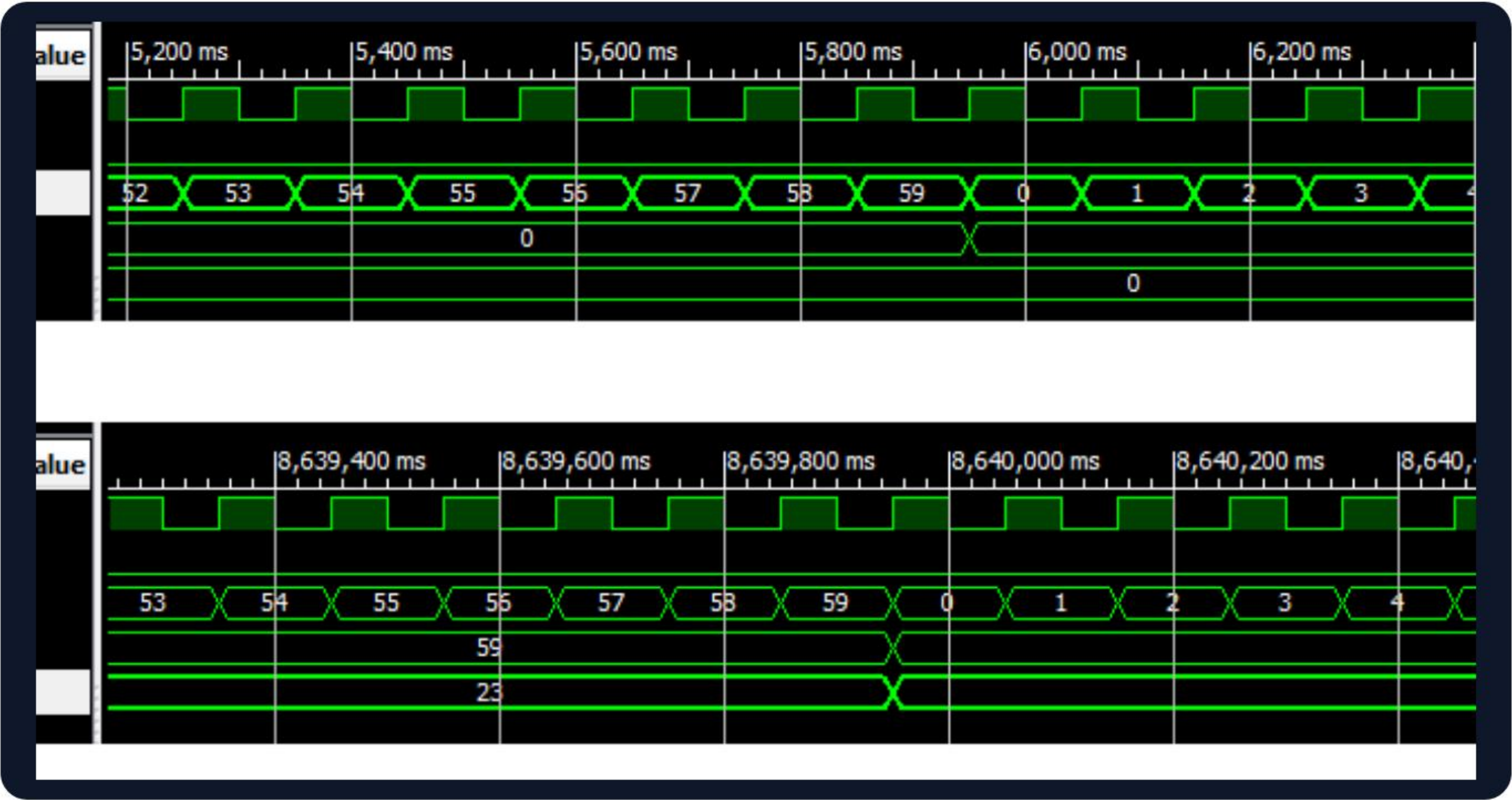
125 MHz

Max Frequency

VERIFICATION

Simulation & Testbench Waveforms

Timing verification under simulated vehicle arrival scenarios



Pulse Verification

Confirms state shift on sensor positive edge.

Auth Response

Valid passcode asserts gate signal within 1 clock cycle.

Boundary Lock Test

Rejects 9th entry request when capacity equals 8.

Hardware Implementation & FPGA Testing

Validation on Xilinx FPGA board with physical peripheral sensors

FPGA Configuration

Target Board: Xilinx Basys 3 / Artix-7.

On-Board LEDs: Displays gate status.

Pmod Headers: Connected to IR modules.

Testing Results

Response Time: Immediate gate motor drive.

Noise Filtering: Debounced sensor input signals.

Accuracy: 100% accurate count over 50 test trials.

PROJECT WRAP-UP

Conclusion & Future Enhancements

Summary of engineering outcomes and potential scalability



Zero State Glitches

Fully synchronous FSM guarantees deterministic gate behavior.



Low Resource Footprint

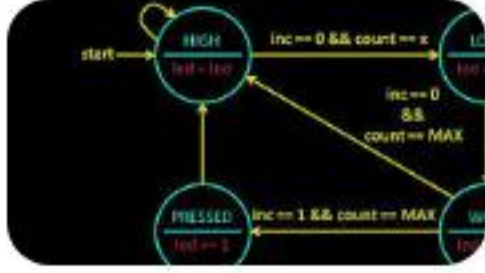
Minimal LUT consumption enables low-cost ASIC implementation.



Future Upgrades

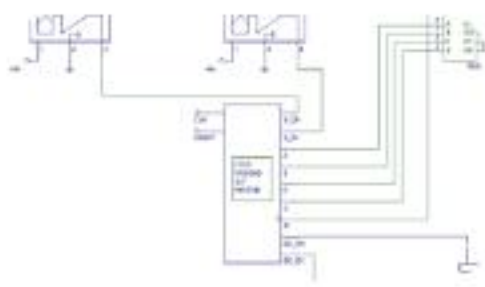
RFID authentication, license plate OCR, and IoT cloud logging.

Image Sources



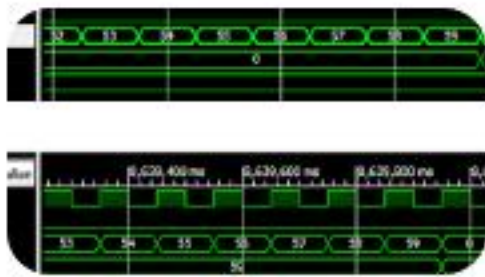
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Source: verilogcodes.blogspot.com